Roll No.: DL.AI.U4AID25009

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – November 2025

First Semester

Artificial Intelligence and Data Science

23EEE103 Introduction to Electrical Engineering

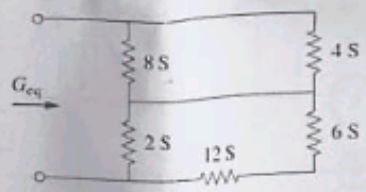
Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

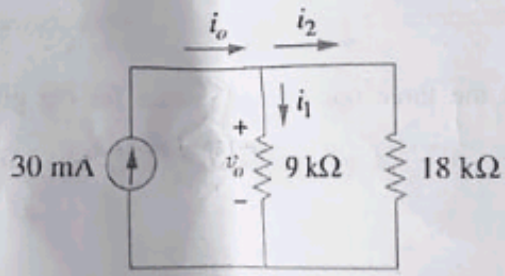
CO	Course Outcomes
CO01	Apply circuit laws and computational techniques to analyse and optimize electrical circuits.
CO02	Simulate and analyse the transient behaviour of DC circuits using computational tools.
CO03	Simplify complex circuits and relevant theorems using Thevenin and Norton equivalents, incorporating AI where applicable.
CO04	Solve and optimize state-space models of circuits with computational methods, leveraging AI techniques.

Answer all questions

1) Evaluate equivalent conductance in the following circuit. [3 marks] [CO01] [BT5Level]

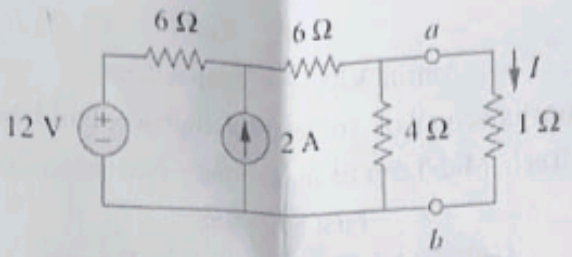


2) Evaluate power supplied and absorbed by each component. [5 marks] [CO02] [BT5Level]



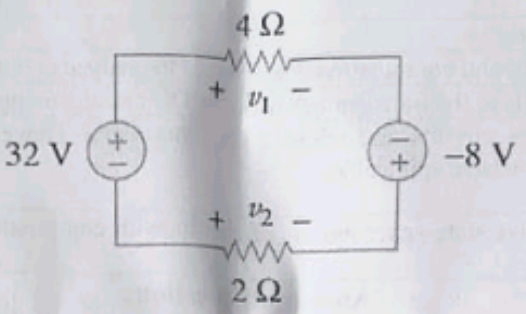
3) Evaluate and draw the Thevenin equivalent circuit to the left of terminal a-b in a given circuit. Also calculate 'I'. [5 marks] [CO03][BTL5 Level]

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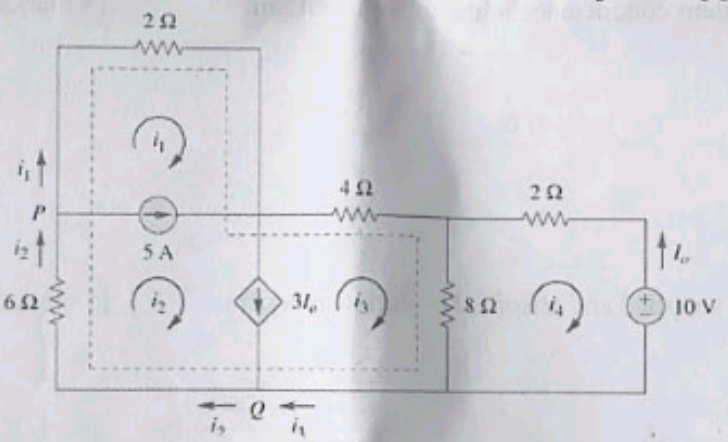


4) A 2.5-kW toaster takes roughly 2.5 minutes to hea eight slices of bread. Find the cost (in US dollars) of operating the toaster once per day for 1 month (30 days). Assume energy costs 8.2 cents/kWh. [3 marks] [CO01] [BT4Level]

5) Find v1 and v2 in the given circuit. [5 marks] [CO02] [BT4Level]

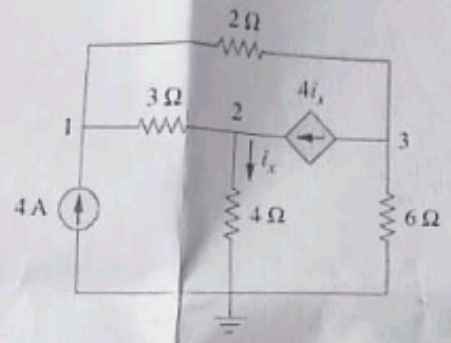


6) Apply mesh analysis in the given circuit. [8 marks] [CO02] [BT3Level]

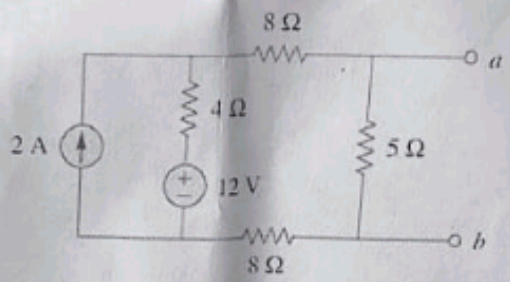


7) Find node voltages at the three non-reference nodes for the given circuit. [8 marks] [CO02] [BT3Level]

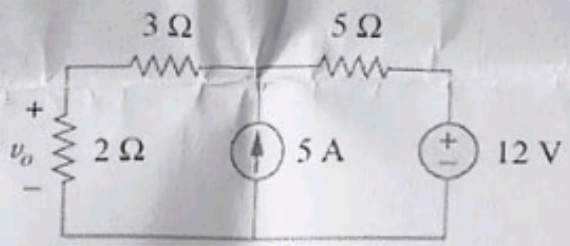
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8) Find Thevenin equivalent and Norton equivalent for the given circuit. *[8 marks] [CO01] [BTL5]*



9) Apply Superposition theorem to find v0. *[5 marks] [CO03] [BTL3]*



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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**14**	BTL 1	-
CO02	**26**	BTL 2	-
CO03	**10**	BTL 3	**21**
CO04	-	BTL 4	**8**
CO05	-	BTL 5	**21**
CO06	-	BTL 6	-

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